Expressions and operator order

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

One connected expression-and-division record.

Explain the model, predict grouped outputs, diagnose an operation error and state a domain limit.

Keep one habit: Keep exact expression, grouping, units/domain, predicted result and observed output/error together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

Values variables and types

My remembered idea, example and correction

MAIN LESSON · SECTION 5

W1 · Explain the procedure

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Packing model: 3 packs*8 each+2 loose=26; 3*(8+2)=30 adds loose to every pack. Separate dimensionless picture 3+2*5=13 versus(3+2)*5=25; multiplier 4 gives 11 and 20. W2: 16-5-3=8,16-(5-3)=14,18/3*2=12.0. Nonnegative 17 in groups 5 gives quotient 3 remainder 2, with ordinary ratio 3.4. Group size 0 is invalid; check each operation separately.

- Write an **expression** for three packs of eight counters plus two loose counters. Explain why adding two within the **parentheses** for every pack describes a different situation.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Predict, trace and observe

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Packing model: 3 packs*8 each+2 loose=26; 3*(8+2)=30 adds loose to every pack. Separate dimensionless picture 3+2*5=13 versus(3+2)*5=25; multiplier 4 gives 11 and 20. W2: 16-5-3=8,16-(5-3)=14,18/3*2=12.0. Nonnegative 17 in groups 5 gives quotient 3 remainder 2, with ordinary ratio 3.4. Group size 0 is invalid; check each operation separately.

- Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Find the first error

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Packing model: 3 packs*8 each+2 loose=26; 3*(8+2)=30 adds loose to every pack. Separate dimensionless picture 3+2*5=13 versus(3+2)*5=25; multiplier 4 gives 11 and 20. W2: 16-5-3=8,16-(5-3)=14,18/3*2=12.0. Nonnegative 17 in groups 5 gives quotient 3 remainder 2, with ordinary ratio 3.4. Group size 0 is invalid; check each operation separately.

- For 17 counters in groups of 5, explain /, // and %. Show a multiplication-and-addition check for the complete groups and **remainder**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · State an honest limit

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Packing model: 3 packs*8 each+2 loose=26; 3*(8+2)=30 adds loose to every pack. Separate dimensionless picture 3+2*5=13 versus(3+2)*5=25; multiplier 4 gives 11 and 20. W2: 16-5-3=8,16-(5-3)=14,18/3*2=12.0. Nonnegative 17 in groups 5 gives quotient 3 remainder 2, with ordinary ratio 3.4. Group size 0 is invalid; check each operation separately.

- Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a **remainder** describe the packing task.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected independent task

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Predict each line. Explain which calculation groups addition before division. What would (17+5)%5 return, and how do /, // and % differ for the supplied positive integers?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Group an expression deliberately

```
print((17 + 5) // 5)
print(17 + 5 // 5)
print(17 / 5, 17 // 5, 17 % 5)
```

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep code and evidence

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Add a labelled **expression**s and **operator** order **evidence** sheet to Python readiness portfolio. Show how you can evaluate numeric **expression**s using python's grouping rules.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use W2: Predict then test **expression**s for **Expression**s and **operator** order. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use the worked case for **Expression**s and **operator** order. Proposed explanation: “All operations are read left to right regardless of **precedence**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Additional practice · Earlier W4 transfer

REFERENCE · SECTION 8

Additional practice · Earlier W4 transfer

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Four trays hold six tiles each and five tiles are loose. Write and evaluate the total **expression**. Then pack the total into complete groups of four and find the **remainder**. Explain why 4 * (6 + 5) is wrong for this story.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- **Parentheses** change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.
- Python 3: print(3+2*5); print((3+2)*5). **Input**s are dimensionless for this arithmetic exercise.
- Trace the operation order in both **expression**s. Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

  Use actual Python 3 results to check the **prediction** after reasoning first.

AI ACADEMY / PYTHON-BRIDGE/3

Expressions and operator order

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Use precedence

In 3+2*5, multiplication happens first.

↓

2. Compute first result

2*5 is 10; adding 3 gives 13.

↓

3. Compute grouped result

Parentheses give 3+2=5 first; multiplying by 5 gives 25.

CHANGE ONE THING

Replace the final multiplier 5 with 4 in both expressions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 13

Reference · Project portfolio milestone

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Project: Python readiness portfolio
- Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.
- For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 14

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- **Parentheses** change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.
- Which labelled step matches this detail? **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.
- 1. Compute grouped result
- 2. Use **precedence**

- 3. Compute first result

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compute grouped result **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. **Parentheses** explicitly group an operation.

Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Changed-case question and hint

- Changed-case practice: Replace the final multiplier 5 with 4 in both **expression**s. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. The first result is 11; the grouped result is 20.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The first result is 11; the grouped result is 20. 3+2*4=3+8; (3+2)*4=5*4.

Repair and keep the record

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print **output**s and explain why they differ.

Does count+tag work, and what happens to each valid **output** after count becomes ten?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Use the picture relevant to a current gap.

01. Expressions and operator order: the case

AI ACADEMY / PYTHON-BRIDGE/3 · 01 · LOOK AT THE FACTS

Expressions and operator order: the case

The situation

Parentheses change an expression. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

Supplied evidence

Python 3: print(3+2*5); print((3+2)*5). Inputs are dimensionless for this arithmetic exercise.

Trace expressions and operator order and explain the changed program before running it.

Follow the labelled picture with the matching explanation. Open the vector diagram

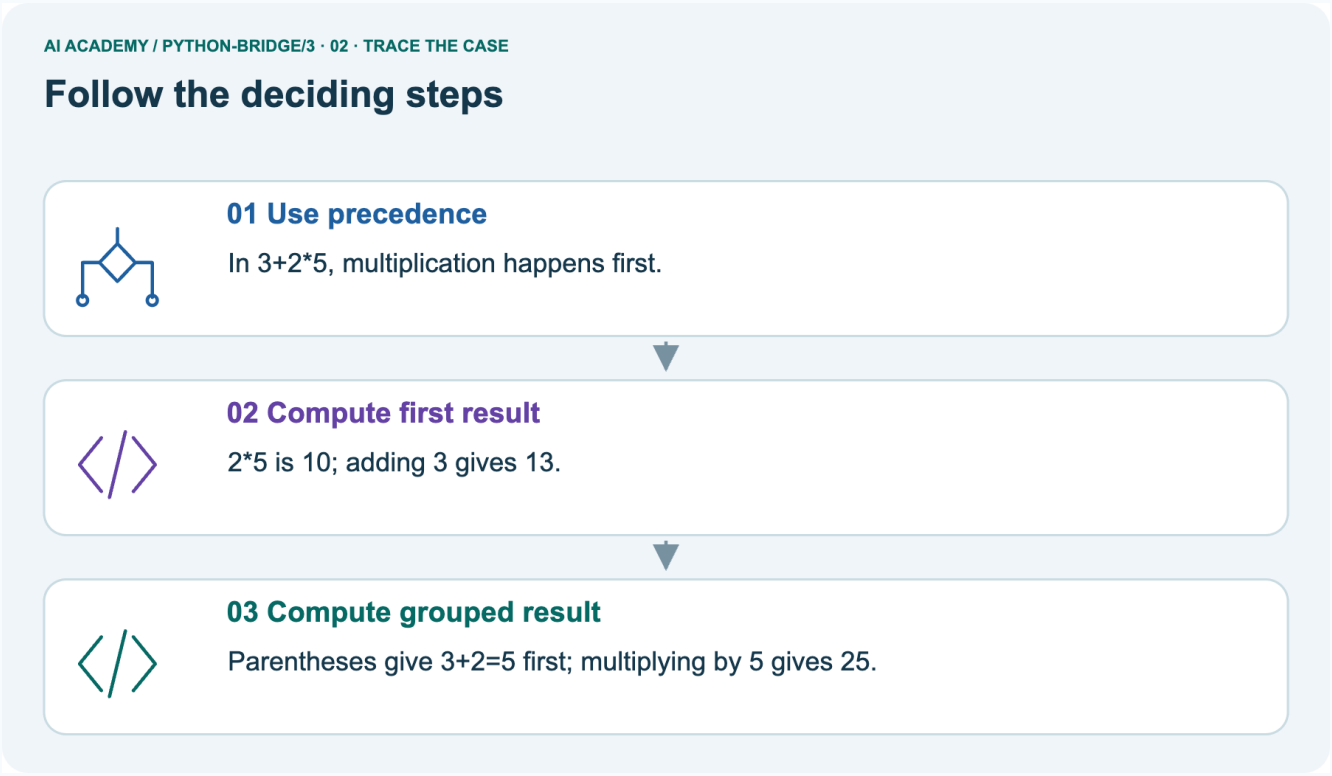
Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group. Supplied facts: Python 3: print(3+2*5); print((3+2)*5).

Inputs are dimensionless for this arithmetic exercise.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



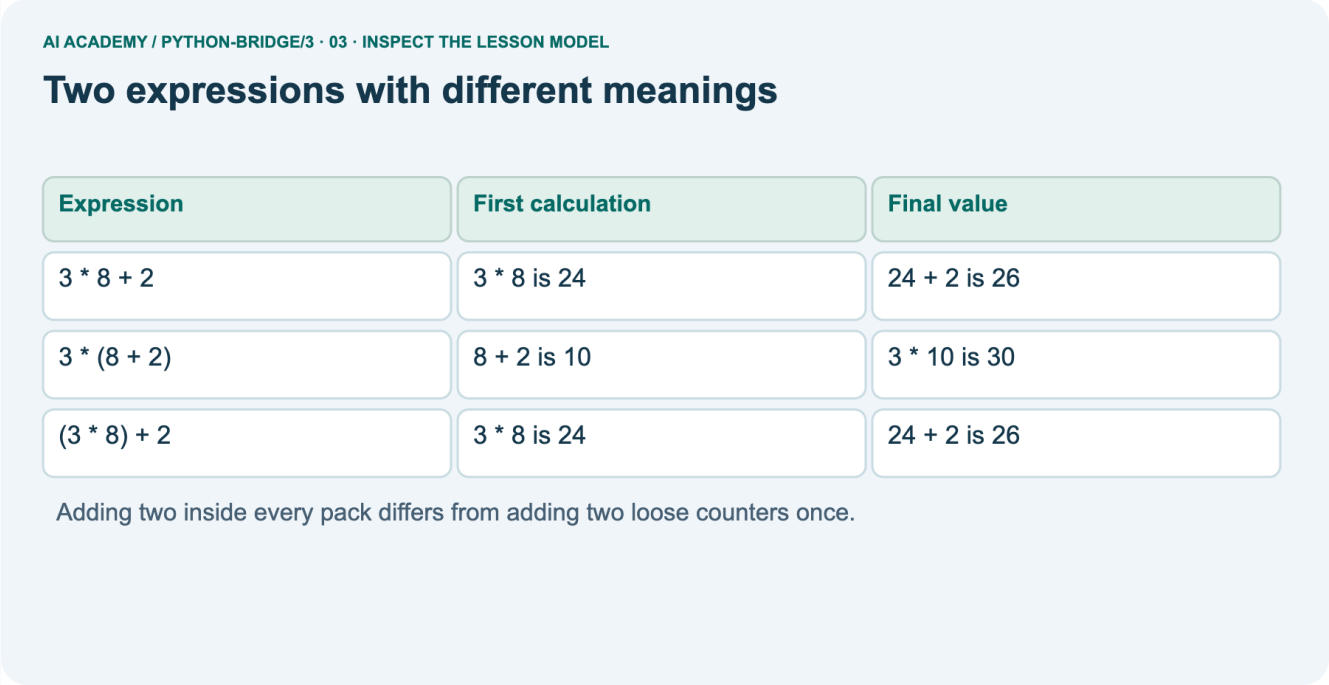
Follow the labelled picture with the matching explanation. Open the vector diagram

Use **precedence**: In 3+2*5, multiplication happens first. → Compute first result: 2*5 is 10; adding 3 gives 13. → Compute grouped result: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two expressions with different meanings



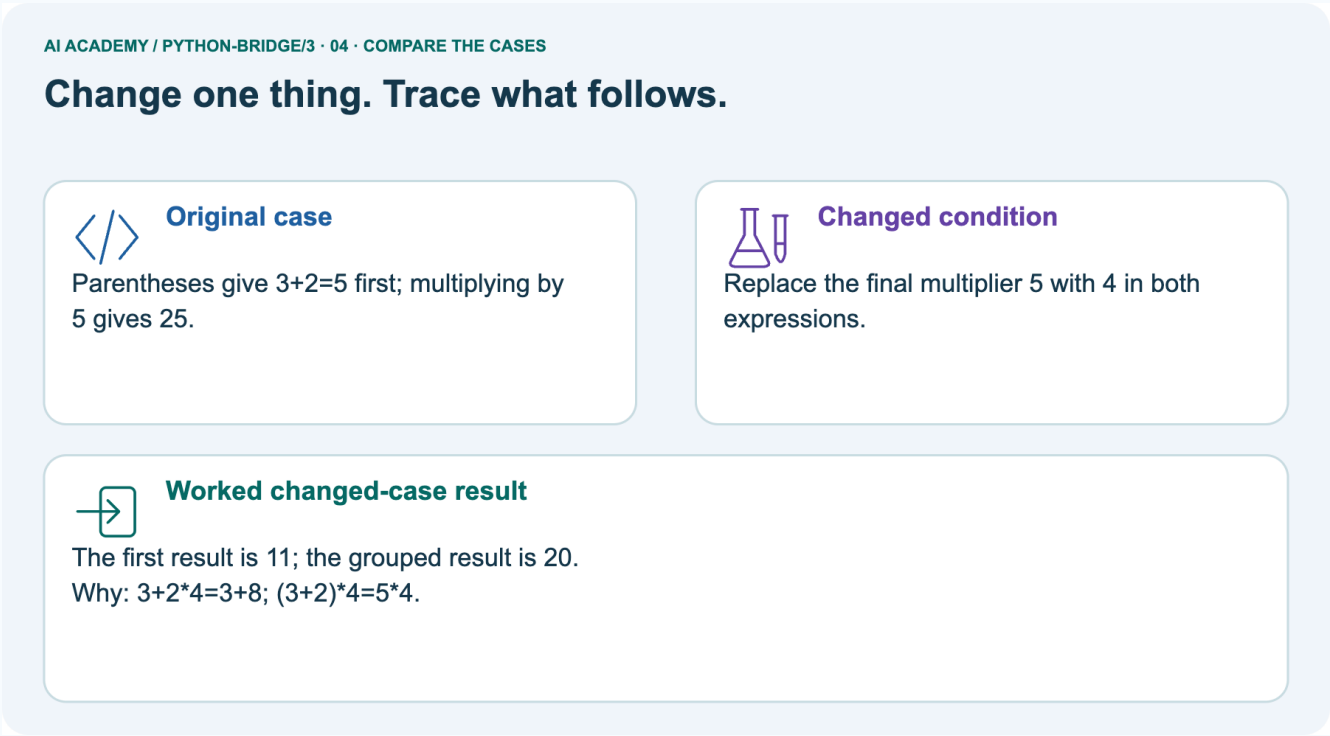
Follow the labelled picture with the matching explanation. Open the vector diagram

Two **expression**s with different meanings. 3 * 8 + 2 | 3 * 8 is 24 | 24 + 2 is 26. 3 * (8 + 2) | 8 + 2 is 10 | 3 * 10 is 30. (3 * 8) + 2 | 3 * 8 is 24 | 24 + 2 is 26

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

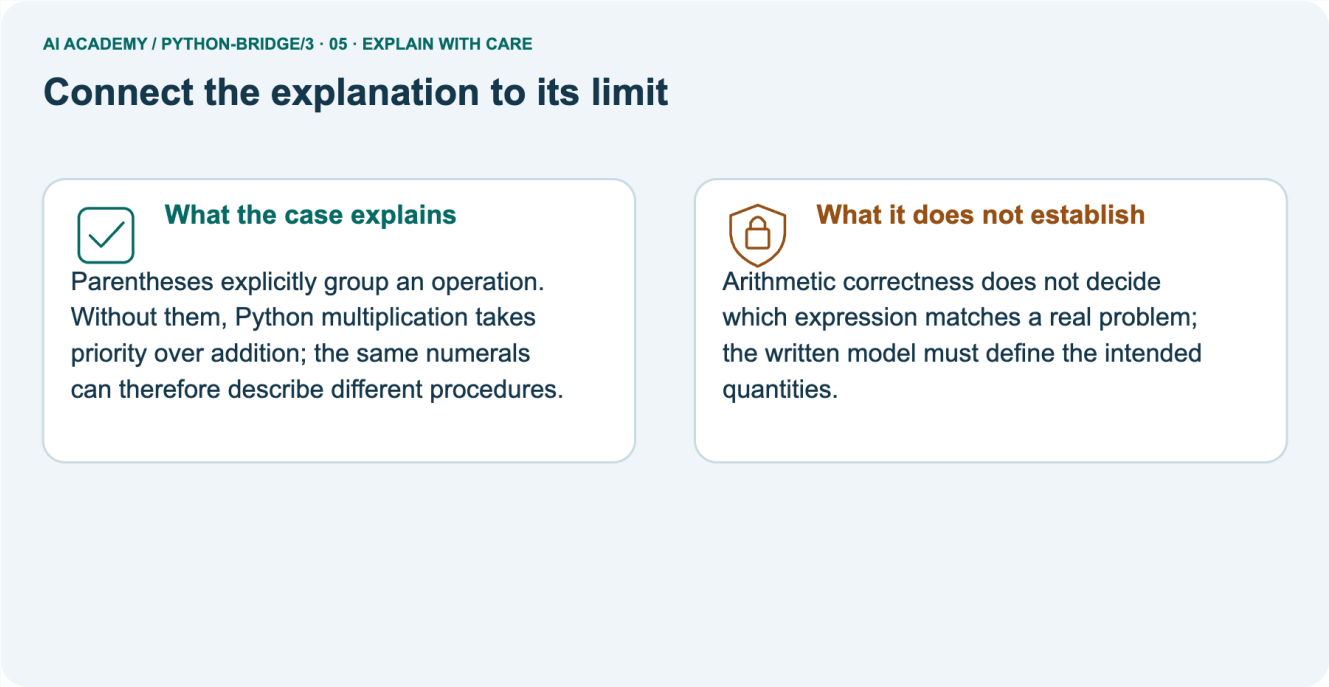
Original case: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. Changed condition: Replace the final multiplier 5 with 4 in both **expression**s. Expected result: The first result is 11; the grouped result is 20.

Reason: 3+2*4=3+8; (3+2)*4=5*4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

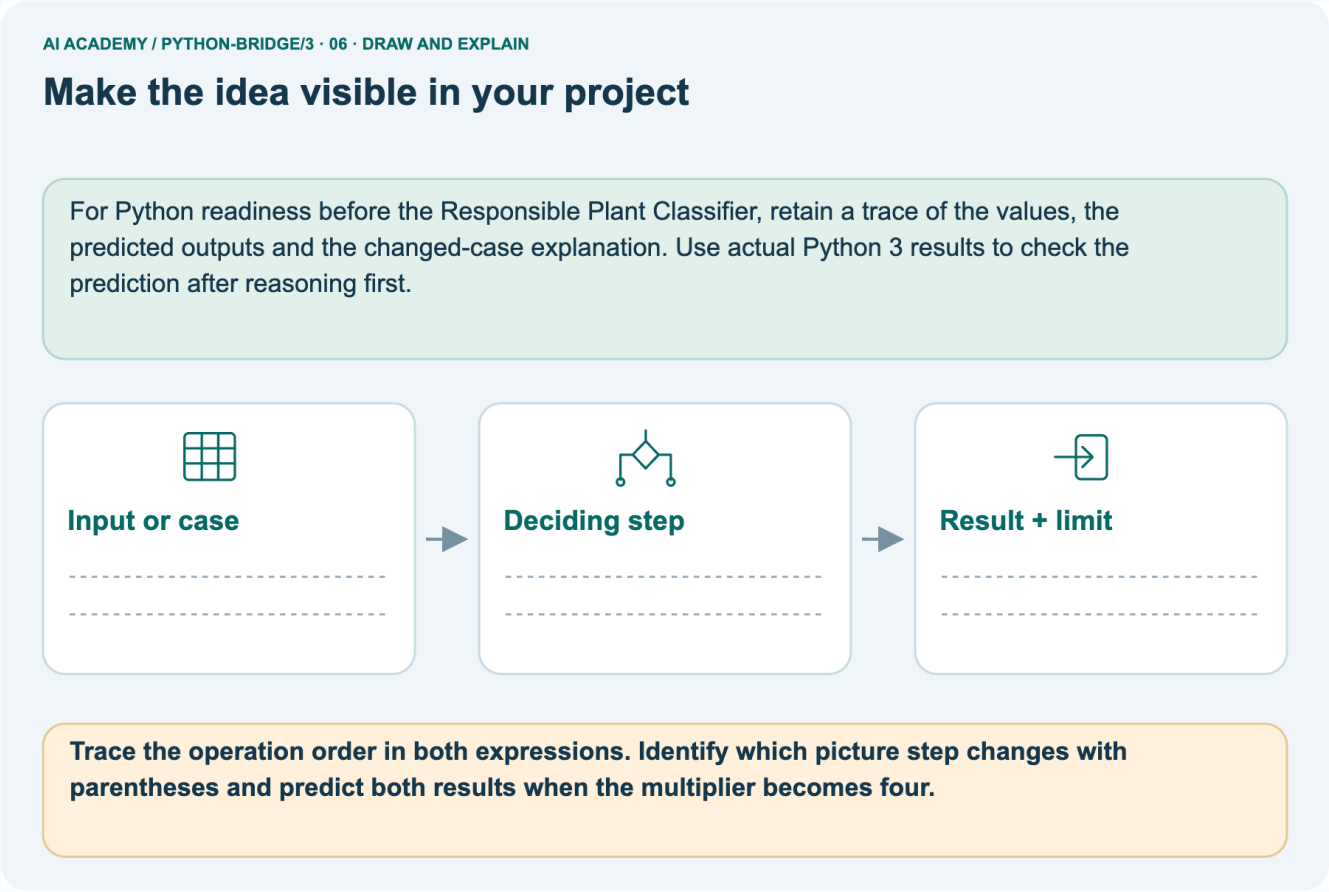
Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Limit: Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Trace the operation order in both **expression**s.

Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Expressions and operator order: the case

02. Follow the deciding steps

03. Two expressions with different meanings

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.